

Energy-Aware and Round Robin Task Scheduling in Cloud Computing: A Comparative Simulation Study

Ahmadreza Hadipour

Department of Computer Engineering

Azad University, Isfahan (Khorasgan) Branch

Isfahan, Iran

arhp.9618@gmail.com

Abstract—Task scheduling is a critical component of cloud computing environments because it directly influences system performance, resource utilization, and energy consumption. This paper presents a comparative simulation study of a traditional Round Robin scheduling algorithm and a custom Energy-Aware scheduling approach. A cloud computing simulation environment was implemented in Java using virtual machines and computational tasks to evaluate the behavior of both scheduling strategies. Performance was assessed using total energy consumption, total workload, and makespan. Experimental results showed that both algorithms successfully processed the assigned workload while exhibiting different performance characteristics. Round Robin achieved a slightly lower makespan and total energy consumption in the current implementation. The study demonstrates the feasibility of incorporating energy-related metrics into scheduling decisions and provides a foundation for future research on intelligent and energy-efficient cloud resource management.

Keywords—Cloud Computing, Task Scheduling, Energy-Aware Scheduling, Round Robin, Virtual Machines, Resource Management

I. INTRODUCTION

Cloud computing has become one of the most important paradigms for delivering computing resources and services over the Internet. Modern cloud environments provide scalable and flexible infrastructure that enables organizations to efficiently manage computational workloads. As the number of cloud users and applications continues to increase, efficient resource management has become a critical challenge.

Task scheduling plays a significant role in cloud computing because it determines how computational tasks are assigned to available virtual machines (VMs). An effective scheduling algorithm can improve system performance, increase resource utilization, and reduce operational costs. Traditional scheduling algorithms such as Round Robin are widely used due to their simplicity and fairness. However, they do not consider factors such as workload distribution or energy consumption.

In recent years, energy efficiency has become an important concern in cloud data centers. Large-scale cloud infrastructures consume significant amounts of electrical power, resulting in increased operational expenses and environmental impact. Consequently, researchers have investigated energy-aware scheduling techniques that

attempt to allocate tasks more intelligently while considering system efficiency.

This study presents a comparative simulation of a traditional Round Robin scheduler and a custom Energy-Aware scheduling algorithm. The proposed system was implemented in Java and evaluated using a cloud simulation environment consisting of multiple virtual machines and computational tasks. Performance was assessed using metrics including total energy consumption, workload distribution, and makespan.

The objective of this project is to investigate how energy-related considerations can be incorporated into scheduling decisions and to evaluate their impact on cloud resource management. The results provide insights into scheduling behavior and establish a foundation for future research in energy-efficient cloud computing.

II. RELATED WORK

Task scheduling has been extensively studied in cloud computing environments. Traditional scheduling algorithms such as First-Come First-Served (FCFS) and Round Robin are commonly used because of their simplicity and low computational overhead. Among these approaches, Round Robin remains one of the most popular scheduling techniques due to its fairness in distributing tasks among available resources.

Despite its advantages, Round Robin scheduling does not consider resource heterogeneity, workload characteristics, or energy consumption. As cloud infrastructures have grown in size and complexity, researchers have proposed more advanced scheduling approaches that incorporate performance and efficiency metrics into task allocation decisions.

Energy-aware scheduling algorithms aim to reduce power consumption while maintaining acceptable levels of system performance. These approaches typically consider factors such as processor utilization, workload distribution, and resource efficiency when selecting a virtual machine for task execution. Various studies have shown that intelligent scheduling strategies can contribute to improved resource utilization and reduced operational costs in cloud environments.

Although significant progress has been made in this area, designing scheduling algorithms that simultaneously optimize energy efficiency and performance remains an open challenge. This study contributes to the field by implementing and evaluating a custom Energy-Aware scheduling strategy and comparing its performance with the traditional Round Robin algorithm.

In addition to traditional scheduling approaches, several researchers have explored intelligent scheduling mechanisms based on optimization and artificial intelligence techniques. Metaheuristic algorithms such as Genetic Algorithms, Particle Swarm Optimization, and Ant Colony Optimization have been applied to cloud scheduling problems to improve resource utilization and reduce execution costs. These methods often achieve better performance than simple scheduling algorithms but require greater computational complexity.

Recent advances in cloud computing have also encouraged the development of machine learning-based scheduling strategies. Such approaches attempt to predict workload patterns and allocate resources dynamically according to expected demand. Although promising, these techniques often require large datasets and extensive training procedures. Consequently, lightweight scheduling approaches remain relevant for small-scale cloud environments and educational simulations.

III. SYSTEM MODEL

The simulated cloud environment consists of a set of virtual machines and a collection of independent computational tasks. A scheduling component is responsible for assigning tasks to available virtual machines according to the selected scheduling policy.

The simulation contains three heterogeneous virtual machines with different processing capacities and power characteristics. Each virtual machine maintains information regarding its computational capacity, current workload, and energy-related parameters.

Tasks represent independent workloads submitted to the cloud environment. Each task is characterized by a unique identifier and a computational length that determines the amount of processing required for execution.

The scheduler acts as the central decision-making component of the system. Its primary responsibility is to determine which virtual machine should execute each incoming task. Different scheduling strategies produce different workload distributions and performance outcomes.

The overall system architecture consists of three layers: task generation, scheduling, and virtual machine execution. Performance metrics are collected after task allocation to evaluate the effectiveness of the scheduling algorithms.

IV. PROPOSED METHOD

Two scheduling algorithms were evaluated in this study: Round Robin scheduling and a custom Energy-Aware scheduling algorithm.

Round Robin scheduling assigns tasks to virtual machines in a cyclic order. The algorithm does not consider workload characteristics, processing capacity, or energy consumption. Its simplicity makes it a useful baseline for performance comparison.

The proposed Energy-Aware scheduler attempts to incorporate system information into scheduling decisions. Before assigning a task, the scheduler evaluates available virtual machines and selects the most suitable candidate according to workload and energy-related considerations.

The objective of the Energy-Aware approach is to distribute tasks more intelligently than a simple cyclic strategy. By considering system characteristics during task allocation, the scheduler seeks to improve resource utilization and support more efficient cloud operation.

Both algorithms were executed using the same set of virtual machines and tasks to ensure a fair comparison. Their performance was subsequently evaluated using identical metrics and experimental conditions.

V. IMPLEMENTATION

The proposed scheduling system was implemented in Java using an object-oriented design. The implementation consists of several classes that collectively model a simplified cloud computing environment.

The VM class represents virtual machines and stores information related to processing capacity, workload, and energy characteristics. The Task class represents computational workloads that are submitted to the scheduler.

The Scheduler class implements the custom Energy-Aware scheduling strategy. It contains the decision logic used to determine which virtual machine should receive a task. The RoundRobinScheduler class implements the baseline Round Robin algorithm.

The Metrics class calculates performance indicators including total energy consumption, total workload, and makespan. These metrics are used to evaluate scheduling effectiveness and compare algorithm performance.

The Main class serves as the entry point of the simulation. It creates virtual machines, generates tasks, executes both scheduling algorithms, and displays the resulting performance statistics.

Finally, the CSVExporter class exports simulation results to a CSV file. This file is subsequently used for

graph generation and performance visualization through Python-based analysis tools.

The virtual machines used in the simulation are heterogeneous, meaning that they possess different computational capacities and energy characteristics. This reflects real-world cloud environments where servers often have varying hardware configurations and performance capabilities. Heterogeneity introduces additional challenges for scheduling algorithms because task allocation decisions can significantly affect both execution time and resource utilization.

The interaction between tasks, schedulers, and virtual machines forms the basis of the experimental framework. By observing how different scheduling algorithms distribute tasks among available resources, it becomes possible to evaluate their effectiveness and identify potential areas for optimization.

VI. EXPERIMENTAL SETUP

The simulation environment consisted of three virtual machines with heterogeneous processing capacities and power characteristics. A set of ten independent computational tasks was generated and submitted to the scheduling system.

Two scheduling algorithms were evaluated:

1. Round Robin Scheduling
2. Energy-Aware Scheduling

Both algorithms were executed under identical conditions to ensure a fair comparison. The same virtual machine configuration and task set were used throughout the experiments.

Performance was evaluated using three commonly used metrics:

- **Total Energy Consumption:** Represents the overall energy used by the virtual machines during task execution.
- **Total Load:** Represents the total workload processed by the cloud environment.
- **Makespan:** Represents the completion time of the final task and serves as an indicator of overall scheduling efficiency.

Simulation results were exported to a CSV file and analyzed using Python-based visualization tools.

Object-oriented programming principles were used throughout the implementation process to improve modularity and maintainability. Each component of the system was implemented as a separate class with clearly defined responsibilities. This design approach simplifies future modifications and allows additional scheduling

algorithms to be integrated without major changes to the overall architecture.

Furthermore, the implementation supports automated result generation through CSV export functionality. This feature enables performance metrics to be analyzed using external tools such as Python and facilitates the creation of graphical representations for research and reporting purposes.

VII. RESULTS AND DISCUSSION

Table I presents the performance results obtained from the simulation.

TABLE I
PERFORMANCE COMPARISON OF SCHEDULING ALGORITHMS

Algorithm	Total Energy	Total Load	Makespan
Round Robin	868575.00	3750	1500
Energy Aware	871533.33	3750	1550

The results indicate that both scheduling algorithms successfully processed the same workload. As expected, the total load remained identical because both approaches executed all tasks generated during the experiment.

The Round Robin scheduler achieved a makespan of 1500, while the Energy-Aware scheduler produced a makespan of 1550. Similarly, total energy consumption for Round Robin was measured at 868575 units compared to 871533.33 units for the Energy-Aware scheduler.

Figure 1 illustrates the comparison of total energy consumption between the two scheduling approaches.

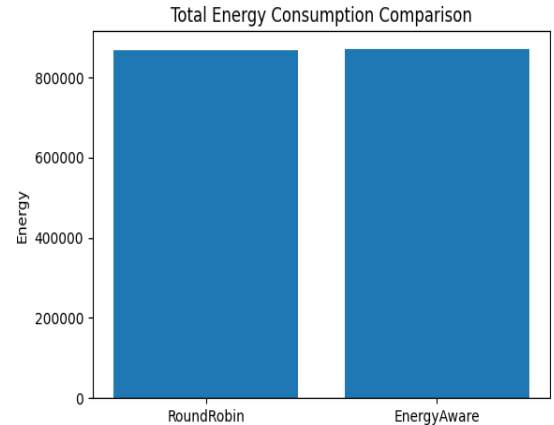


Fig. 1. Total energy consumption comparison between Round Robin and Energy-Aware scheduling.

The energy comparison shown in Fig. 1 demonstrates that both scheduling algorithms produced similar energy consumption levels. The difference between the two approaches was relatively small under the current experimental conditions.

Figure 2 presents the makespan comparison obtained during the experiments.

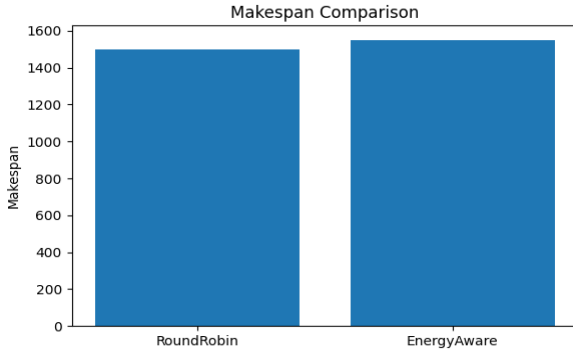


Fig. 2. Makespan comparison between Round Robin and Energy-Aware scheduling.

As illustrated in Fig. 2, the Round Robin scheduler completed the workload slightly faster than the Energy-Aware scheduler. This result suggests that additional optimization of the scheduling policy may be required to improve execution efficiency.

Figure 3 shows the total workload processed by each algorithm.

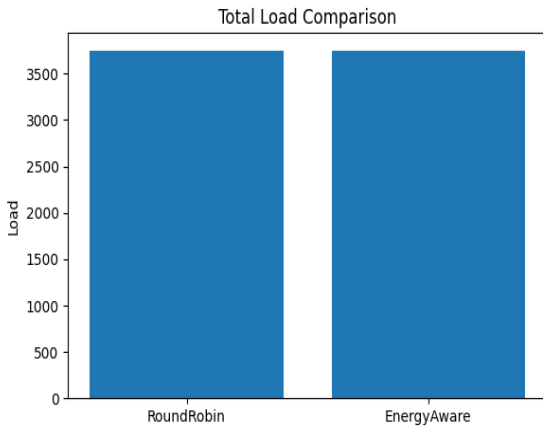


Fig. 3. Total workload processed by both scheduling algorithms.

The workload comparison confirms that both algorithms processed the same amount of computational work. Therefore, the observed differences in performance are attributable to scheduling decisions rather than differences in workload size.

Overall, the experimental findings demonstrate that the proposed Energy-Aware scheduling strategy produced performance comparable to Round Robin in the current implementation. Although the Energy-Aware scheduler did not outperform the baseline algorithm, the study demonstrates the feasibility of incorporating energy-related considerations into scheduling decisions. Future improvements may focus on refining the scheduling cost function, improving workload balancing, and evaluating larger cloud environments.

VIII. LIMITATIONS AND FUTURE WORK

The current implementation represents a simplified cloud computing environment and therefore has several limitations. First, the simulation includes a relatively small number of virtual machines and tasks. Larger experimental environments may produce different scheduling behaviors and provide additional insights into algorithm scalability.

Second, the Energy-Aware scheduler uses a basic decision-making strategy that does not incorporate predictive analytics or workload forecasting. More advanced approaches may achieve better performance by considering historical execution patterns and dynamic system conditions.

Future work will focus on extending the simulation to larger cloud infrastructures, integrating machine learning techniques for intelligent scheduling, and evaluating additional performance metrics such as throughput, response time, and resource utilization efficiency.

IX. CONCLUSION

This paper presented a comparative simulation study of Round Robin and Energy-Aware scheduling algorithms in a cloud computing environment. A Java-based simulation framework was developed to model task allocation among multiple virtual machines and evaluate scheduling performance.

Experimental evaluation was performed using total energy consumption, total workload, and makespan as performance metrics. The results showed that both algorithms successfully processed the workload while exhibiting different scheduling characteristics.

Although the Energy-Aware scheduler did not achieve lower energy consumption or makespan in the current configuration, the project demonstrates how energy-related metrics can be integrated into scheduling decisions. The implementation provides a foundation for future experimentation and optimization of intelligent cloud scheduling strategies.

Future work may include dynamic resource scaling, machine learning-based scheduling, workload prediction, and integration with advanced cloud simulation platforms such as CloudSim.

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